

JADE OS · The AI Operating Layer for Minnesota Freight

Six agents. One pane of glass. Verifiable ROI in under 90 days.

slide 1 · title

The Operator's Bleed

- Empty miles still 15-22% of fleet hours · ATRI 2024
- Dispatcher burden: 60+ load decisions per shift, mostly tribal knowledge
- Compliance violations cost \$1,000-\$5,000 per incident · FMCSA
- Driver replacement cost ~\$9.5k each · turnover ~78% mid-market

[slide 2](#) · [problem](#)

JADE OS · Six Agents Composing One Workflow

- Dispatch Optimizer · 8-15% empty mile drop
- Route & Fuel Agent · 10-18% fuel savings
- Compliance Agent · 40-60% violation drop
- Pricing Agent · Rate-floor guard prevents below-cost quotes
- Driver Retention Agent · +6-12pp 12-mo retention
- Predictive Maintenance · -20-30% unplanned downtime

Module 1 · Dispatch Optimizer

- Ingests ELD HOS, TMS open loads, driver prefs
- Constraint solver + utility scoring
- Writes back to TMS on approval (L2 autonomy)
- 60-second decision · 30-50% dispatcher relief

Module 2 · Route + Fuel

- EIA daily diesel by region
- DOT 511 + weather + truck-restricted layer
- Fuel-arbitrage MILP picks cheapest qualifying stop
- Average 10-18% fuel margin recovery

Module 3 · Compliance + Safety

- Rules engine on FMCSA 395/396
- Audit-pack auto-generation
- Pre-trip risk score blocks high-violation dispatches
- Audit prep · days → hours

ROI · Mid-Market 175 Trucks

- Annual savings ~ \$1.8–2.4M
- Upfront \$130k · Annual license \$210k
- Payback ~ 2 months · 3-yr NPV ~ \$4.5M @ 10%
- Conservative band -10%; Optimistic +10%

Pilot Profile · Bay & Bay-class (illustrative)

- Mid-market, intermodal-leaning fleet
- Initial agent · Dispatch + Compliance
- 30-day baseline, 60-day pilot, 90-day decision gate
- Success metrics defined in writing pre-pilot

Pilot Profile · Specialized / Hazmat

- 50-150 trucks, specialized freight
- Lead with Compliance Agent · highest pain
- Add Route+Pricing after compliance ROI proven

slide 9 · case

Why JADE OS vs Descartes / Omnitracs / Geotab

- Vendor-agnostic AI layer ON TOP of your TMS, not a rip-and-replace
- Rate-floor & audit trails are FIRST-class, not after-thoughts
- Six agents share workflow memory · one auditable thread
- MN-grounded benchmarks; not coast-trained mocks

Pricing & Pilot Structure

- 6-month pilot · \$25-40k all-in (subsidized integration)
- Annual license post-pilot · \$1,200/truck typical
- Decision checkpoints at month 4
- Pilot exit · roll-forward credit if you convert

The Ask

- 30-min discovery this week
- 1-day data-readiness assessment
- Pilot agreement within 14 days for hot-list carriers